

KIT 2

KIQ_2_1

Analytical commentary — Top Technology Domains (IPC)

Headline. The IPC distribution maps onto a coherent digital ordering technology stack: commerce logic (G06Q), computing infrastructure (G06F), and intelligence (G06N) form a dominant top tier (~3,300 combined IPC occurrences), with a second tier of perception and communication technologies (image, video, transmission) at 100–155 occurrences each, and a long tail of specialized capabilities — voice (G10L), IoT (G16Y), wireless (H04W), QR/OCR coding (G06K) — that signal where the field is expanding next.

The stack reads top-down:

G06Q (1300) — commerce logic. Business methods is the largest single domain. Every patent in this corpus is fundamentally about transactions, payments, recommendations, pricing, marketplaces. This is the layer that touches METRO's existing business directly.

G06F (1237) — computing infrastructure. Almost tied with G06Q. This is the plumbing — backends, databases, software architectures, cloud systems. It tells you these aren't theoretical patents; they're implementation-grade inventions designed to run in production.

G06N (761) — AI / machine learning. Roughly 60% of the commerce-layer volume. This is the early-warning signal: a substantial share of digital ordering patents now embed AI/ML as a core component, not as an add-on. Three years ago this would have been a fraction of what G06Q represents; today it's a co-dominant layer.

H04N, G06V, H04L, G06T (100–155 each) — perception & transmission. Video, image recognition, image processing, digital transmission. Technologies for product visualization, automated catalog tagging, AR-assisted ordering, real-time data flows. Mid-volume but accelerating.

G10L (61), G16Y (19) — voice and IoT. Smaller volumes but strategically the most informative

for KIT 2 — these are the new interaction modes (voice ordering in HoReCa kitchens, IoT-enabled autonomous replenishment for high-turn supplies). Low counts now, but their presence in a 2022–2026 corpus is the leading indicator.

How this connects to METRO and your project

The three-layer story for the report:

"The IPC distribution confirms that emerging digital ordering technologies are not a single innovation but a layered stack: commerce logic (G06Q), computing infrastructure (G06F), and AI (G06N) form the dominant foundation, complemented by perception technologies (image, video) and emerging interaction modes (voice, IoT). For METRO, this means the digital ordering channels of the next 3–5 years will not be re-platformed once; they will be progressively augmented across the stack. Strategic deployment should therefore prioritize the layers METRO controls — commerce logic and customer interaction — and partner upstream for the infrastructure and AI layers, where filing activity is dominated by external players (Baidu, Tencent, IBM, Strong Force portfolios)."

Tying it to KIQ 1.3 explicitly:

In KIQ 1.3 you concluded that AI, ML, CRM, marketplaces sit at the technology periphery of the retention discourse — enablers, not drivers. KIQ 2.1 zooms into that periphery and shows: (1) it's accelerating (the trend chart), (2) it's being built by external technology vendors (the applicant chart), (3) it spans a full stack from infrastructure to user interface (this IPC chart). The two KIQs answer different questions but tell one connected story:

Retention is delivered by satisfaction, trust, and experience. The technologies that amplify those mechanisms are accelerating, externally controlled, and span a multi-layer stack. METRO's strategic challenge for the next 3–5 years is not whether to adopt these technologies, but which layers to build in-house, which to partner for, and which to acquire.

That's the kind of synthesis that turns three descriptive charts into one strategic recommendation — exactly what KIT 2 (early warning) is supposed to produce.

One sentence for the executive summary

"Patent intelligence reveals a layered digital ordering technology stack — commerce, computing, and AI dominate; voice and IoT signal where the next wave of customer interaction is being built — with external vendors holding most of the IP and a clear deployment window still open for European retailers."